

Budget-Aware Adaptive LoRA: Gradient-Driven Rank Allocation for Parameter-Efficient Fine-Tuning

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INTRODUCTION

Large language models (GPT, BERT, LLaMA) are costly in terms of computation, as it is required to update billions of parameter to fine tune the full model. Adaptation of these LLMs for a task specific problem is prohibitively expensive for most researchers and organizations. **LoRA** (Low-Rank Adaptation) reduces the trainable parameters using small adapter matrices, and uniform rank across all layers.

- Traditional fine-tuning of LLMs:

$$W_{\text{new}} = W_{\text{pretrained}} + \Delta W ; (\Delta W \text{ is huge as } W_{\text{pretrained}})$$

- LoRA's approach:

$$W_{\text{new}} = W_{\text{pretrained}} + B \times A ; (B \& A \text{ are very small } W)$$

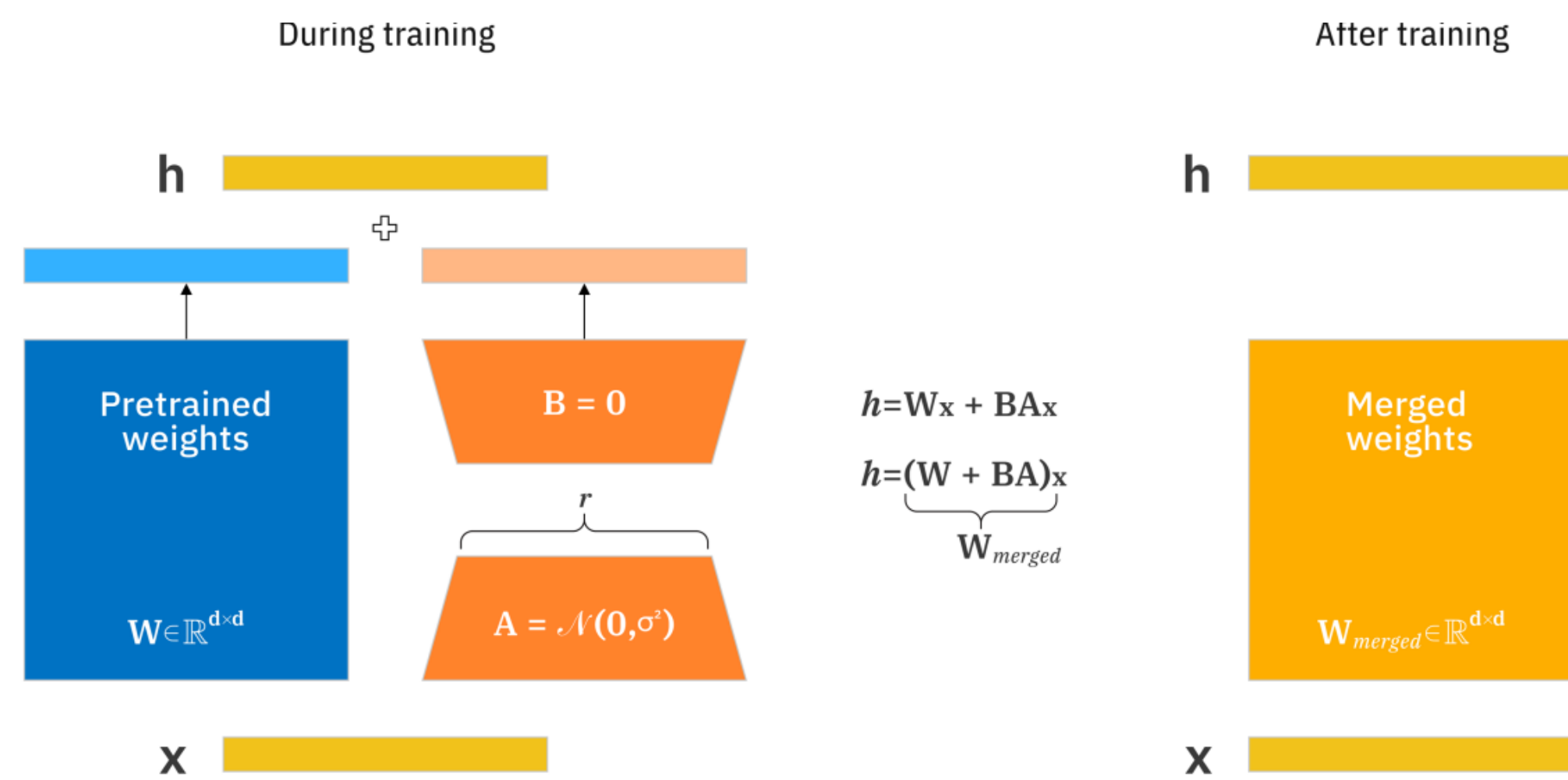


Figure 1: The workflow of LoRA with adapter matrices

LoRA represents weight updates in a low intrinsic dimension. Instead of updating the ($12,288 \times 12,288 = 150M$) parameters, LoRA plays with a smart trick of two tiny matrices $B(12,288 \times 8)$ & $A(8 \times 12,288)$ which turns the billions of trainable parameters into just 196K. Here, 8 is the rank which is uniform across all the layers.

Research Question

Does adaptive rank allocation improve upon uniform LoRA under strict parameter budgets?

LoRA reduces the trainable parameters and GPU usage in a massive number. However, it lacks on some points: The rank allocation is not optional, since it assigns rank uniformly. Finding the right rank needs a grid search over $r \in \{2, 4, 8, 16, 32, 64\}$, which is costly.

BA LoRA's adaptive allocation strategy combines (1) Gradient based layer importance estimation. (2) Budget-aware adaptive rank allocation. (3) Single pass training (comparable efficiency).

Method	Adaptive	Budget	Single Pass
LoRA	✗	✓	✓
AdALoRA	✓	✗	✗
GoRA	✓	✗	✓
BA-LoRA	✓	✓	✓

Table 1: Features eligibility of LoRA, ALoRA, GoRA, and BA-LoRA

RESULTS

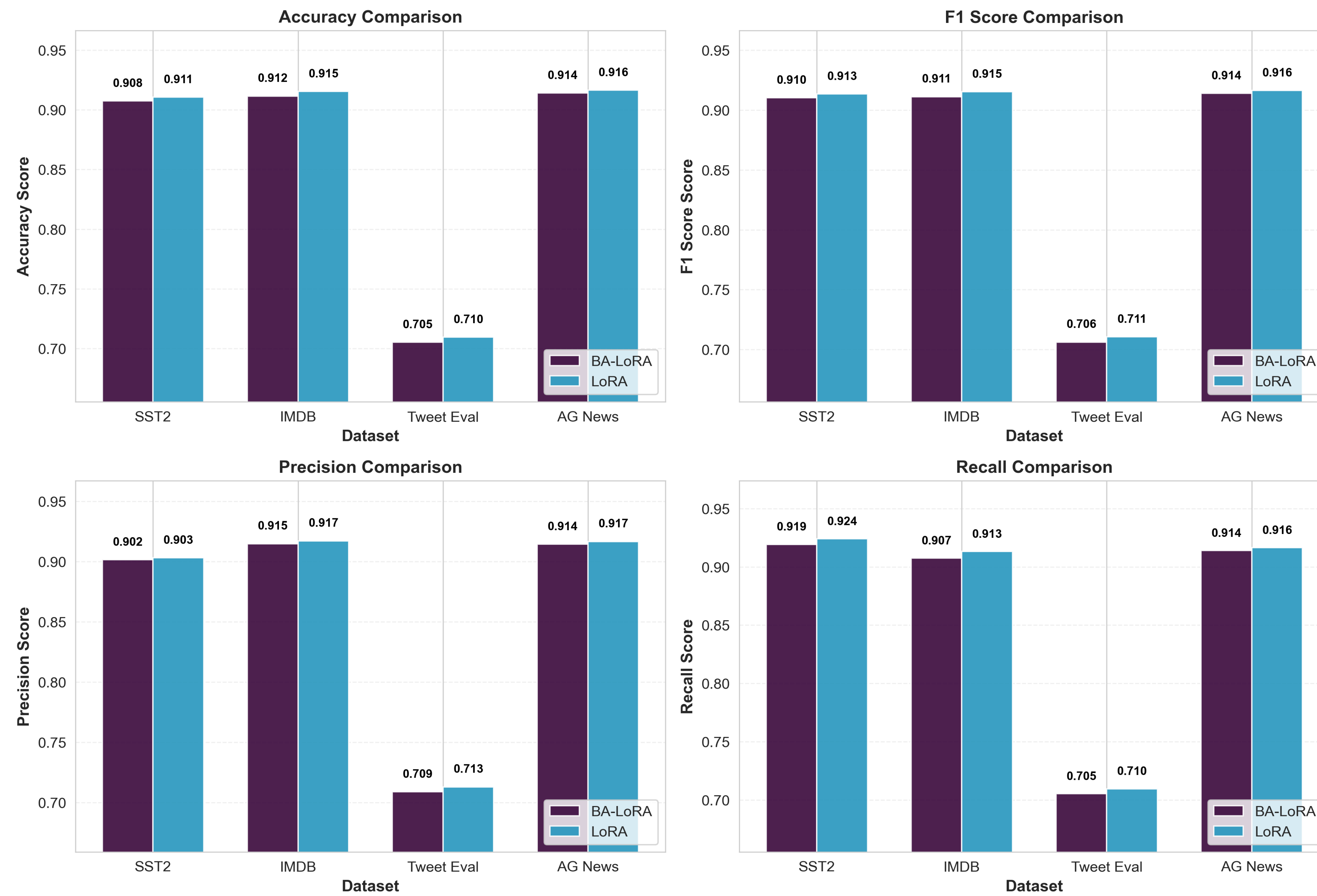


Figure 2: BA-LoRA vs LoRA: Multi-Dataset Performance Comparison

Adaptive rank allocation does NOT significantly improve upon uniform LoRA for text classification with base models. Across 103 experiments spanning 4 datasets and 2 architectures, we found no meaningful performance or efficiency advantages.

Key Observations:

- No Statistical Difference: Welch's t-test ($\alpha=0.05$) shows p-values of 0.38-0.84; accuracy differences within $\pm 0.4\%$.
- Similar Efficiency: Training time comparable on 3/4 datasets; gradient computation overhead ($<5\%$) negligible in practice.
- Consistent Across Scales: Both DistilBERT (66M) and RoBERTa (125M) showed identical patterns, suggesting findings generalize within this model size range.
- Budget Equivalence: Under strict parameter constraints (**630K-815K**), adaptive allocation provides no advantage over uniform distribution.

Dataset	BA-LoRA Acc	LoRA Acc	Δ Acc	p-value	Significant	Time Overhead
SST2	0.9076 ± 0.0255	0.9107 ± 0.0254	-0.0031	0.7634	✗	9.8%
IMDB	0.9116 ± 0.0212	0.9154 ± 0.0200	-0.0038	0.6381	✗	-21.1%
TWEET_EVAL	0.7052 ± 0.0142	0.7095 ± 0.0122	-0.0043	0.4326	✗	-0.7%
AG_NEWS	0.9141 ± 0.0037	0.9165 ± 0.0006	-0.0024	0.3845	✗	-1.2%

Table 2: Statistical summary

METHODS

We evaluated BA-LoRA against vanilla LoRA across 4 datasets and 2 model architectures in 103 experiments.

Datasets & Models: **SST-2** (67K train), **IMDB** (25K train), **AG News** (120K train), **TweetEval** (45K train) • **DistilBERT-base** (66M params) & **RoBERTa-base** (125M params) • Ranks tested: $r \in \{2, 3, 4, 6, 8, 12\}$.

BA-LoRA Pipeline:

- Gradient Importance:** $I(W) = \text{avg}(|W \odot \nabla W|)$ on 5K samples
- Budget-Aware Allocation:** Ranks $\in [0.5\times, 2.0\times]$ base_rank Iterative adjustment $\rightarrow$ exact budget
- Standard Fine-Tuning:** Random init, AdamW, lr=5e-4, epochs=3

Evaluation: Accuracy, F1, Precision, Recall | Statistical testing: Welch's t-test ($\alpha=0.05$)

DISCUSSION

Why Didn't Adaptive Allocation Help?

- Task Simplicity: Classification tasks may not benefit from layer-specific adaptation; decision boundaries learnable with uniform ranks.
- Model Scale: Benefits likely emerge at larger scales (7B+params); base models (66M-125M) may have insufficient layer heterogeneity.
- Metric Limitations: Gradient magnitude may not fully capture layer importance; Fisher information or activation statistics may work better.

Despite comparable performance, this work provides valuable insights:

- Rigorous Framework: Large-scale study (103 exp) with strict budget enforcement; provides reproducible benchmarks.
- Validates Simplicity: Negative results prevent overengineering; uniform LoRA highly effective for classification.
- Practical Guidance: Use uniform LoRA for classification; reserve adaptive methods for LLMs 7B+ or complex reasoning tasks.

References

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